

TFScope: Sequence-Only Prediction of Transcription Factor DNA-Binding Specificity with a Contact-Aware Protein Language Model

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Abstract

The DNA-binding specificity of a transcription factor (TF), summarized as a position weight matrix (PWM), is a foundational quantity in gene regulation, yet most TFs across the proteome have never been experimentally characterized and the leading computational predictors require an experimentally solved or modeled protein–DNA complex at inference. Here we present **TFScope**, a framework that predicts PWMs directly from TF amino-acid sequence, with no structural input. TFScope couples a protein language model encoder (ESM-2, 650 M parameters) with low-rank adaptation, dual-stream gated pooling of the DNA-binding domain (DBD), a family-conditioned mixture-of-experts decoder, and a contact-aware cross-attention head that localizes specificity-determining residues. On the 130-TF blind split of the structure-based state-of-the-art DeepPBS, TFScope exceeds it on 8 of 11 motif-level metrics (mean per-column Pearson r 0.802 vs. 0.750); even with retrieval disabled, the sequence encoder alone already matches DeepPBS (0.749). Under a 40%-identity out-of-distribution benchmark TFScope attains mean $r=0.535$, and leave-family-out analysis exposes a ~ 0.48 sequence-only transfer floor. Cross-attention recovers the canonical zinc-finger and basic-helix–loop–helix recognition codes from sequence alone, and identifies long C2H2 zinc-finger arrays as the prediction frontier. TFScope enables proteome-scale specificity annotation at negligible cost.

22 Introduction

23 The sequence preference of a transcription factor for its genomic binding sites is one of the central
24 determinants of where, when, and how strongly a gene is expressed. Encoded in the physico-chemical
25 interface between a TF’s DNA-binding domain and the DNA double helix [1, 2], this preference is
26 conventionally summarized as a position weight matrix (PWM): a compact, additive model of the
27 per-position base frequencies in bound sites. PWMs underpin nearly every downstream analysis of
28 gene regulation, from scanning genomes for regulatory elements and reconstructing gene regulatory
29 networks, to interpreting the functional consequences of non-coding genetic variation that disrupts
30 or creates binding sites [3, 4]. Because mutations that alter TF binding contribute to developmental
31 disorders and cancer, and because engineered TFs are increasingly used as tools in synthetic biology,
32 accurate, proteome-scale knowledge of binding specificity has both basic and translational value.

33 Despite decades of effort, experimentally determined specificities remain sparse. High-throughput
34 assays such as protein-binding microarrays [5], HT-SELEX [6], and ChIP-seq have characterized only
35 a fraction of the ~1,600 human TFs [7, 8], and coverage is far poorer across the broader eukaryotic
36 proteome and for the many allelic and paralogous variants whose specificities differ subtly but
37 consequentially. Each assay is laborious, organism- and condition-specific, and cannot keep pace
38 with the rate at which new TF sequences are catalogued. A predictive model that maps amino-acid
39 sequence directly to binding specificity would therefore fill a persistent and widening gap.

40 Computational approaches to this problem fall into two broad classes, each with a characteristic
41 limitation. The first encodes hand-derived “recognition codes,” most famously the Cys₂His₂
42 zinc-finger code in which a small set of recognition-helix residues specify the contacted DNA
43 triplet [9, 10]; such codes are accurate within the family they were derived for but do not generalize
44 across the full diversity of TF structural classes. The second class learns directly from functional
45 genomics data [4, 11], but these models are trained per-TF or per-cell-type and presuppose the
46 very experimental data we lack for uncharacterized factors. Most recently, structure-based deep
47 learning has set the state of the art: DeepPBS [12] reads DNA base preferences directly from the
48 heavy-atom contact geometry of a docked protein–DNA complex and, combined with advances
49 in structure prediction [13], can in principle be applied proteome-wide. In practice, however, the
50 structure requirement is a fundamental bottleneck: accurate docked complexes are unavailable

for most TF–DNA systems, modeling each complex is computationally expensive, and prediction quality degrades when the input structure is itself a prediction rather than a crystal.

Protein language models (PLMs) offer a route around the structure bottleneck. Trained on hundreds of millions of natural sequences, models such as ESM-2 [14] capture evolutionary, structural, and functional regularities that transfer to diverse downstream tasks [15]. Crucially, the specificity of a TF is governed by a small number of base-contacting residues in its DBD [10, 16], and these residues leave detectable imprints in co-evolutionary and language-model representations. We hypothesized that a PLM encoder, fine-tuned on the DBD with low-rank adaptation [17], would expose enough of this signal to predict binding specificity from sequence alone, without recourse to any three-dimensional structure.

Here we introduce TFScope, a sequence-only framework for PWM prediction. TFScope encodes the TF through a PLM backbone, pools DBD-focused and global representations, decodes through a family-conditioned mixture-of-experts layer [18], and emits a gated PWM through a contact-aware cross-attention head that learns to attend to specificity-determining residues. An optional retrieval branch supplies nearest-neighbor PWMs from experimentally characterized TFs, providing an explicit, learned analogue of the homology transfer long used informally in structural bioinformatics. We benchmark TFScope against the structure-based state of the art under three regimes of increasing stringency, dissect its behavior on well-characterized factors to show that it recovers canonical recognition codes from sequence alone, and delineate honestly where it generalizes and where it does not.

Results

A sequence-only framework for binding-specificity prediction

TFScope maps a TF amino-acid sequence to a $4 \times L$ PWM through five stages (Fig. 1). The *encoder* is ESM-2 (650 M parameters, frozen) [14] with LoRA adapters (rank 16, α 32) on its last six transformer layers (~491 K trainable parameters); a learned DBD indicator embedding marks residues inside the annotated domain, biasing representation toward the recognition interface. *Dual-stream gated pooling* summarizes the sequence with two attention-pooled vectors — one over the whole protein, one restricted to the DBD — which are concatenated and projected to a

512-dimensional joint representation. A *family-conditioned mixture-of-experts (MoE) decoder* [18] (12 routed plus 1 shared expert, top-2 routing) is conditioned on semantic family embeddings derived from PLM encodings of family-level descriptions, so that even a structurally unfamiliar TF receives a meaningful prior without any label leakage. A *contact-aware cross-attention head* maps PWM-column queries onto DBD residues; a residual branch ($z = z_{\text{prior}} + \lambda \Delta z_{\text{contact}}$) with cosine attention and row-diversity and hub penalties adds 0.55 M trainable parameters while leaving the rest of the model frozen. Finally, a *position gate* selects information-rich columns and a regression head emits per-column base frequencies; an optional retrieval branch injects $K=3$ nearest-neighbor PWMs, indexed with leave-gene-out (LGO) hygiene, before the decoder.

Sequence-only prediction rivals structure-based specificity models

We compared TFScope with DeepPBS [12] on the latter’s 130-TF blind test split, applying two evaluation protocols identically to both methods. The *oracle-aligned* protocol trims each prediction to the target’s information-rich core (per-position information content, $\text{IC} \geq 0.25$) and grants oracle offset and reverse-complement alignment, isolating base-composition accuracy from placement error. The *canonical-fixed* protocol independently canonicalizes prediction and target and grants no alignment, reflecting deployment.

Under oracle alignment, TFScope exceeded DeepPBS on 8 of 11 metrics across the 116 TFs both methods cover (Table 1). Mean per-column Pearson r rose from 0.750 to **0.802**, median r from 0.759 to **0.831**, top-1 base accuracy from 0.793 to **0.836**, Matthews correlation from 0.698 to **0.765**, and mean absolute error fell from 0.133 to **0.108**; DeepPBS retained an edge only on the two calibration metrics (cross-entropy and KL divergence). Disabling the contact-aware branch — leaving only the frozen prior (–contact) — dropped mean r to 0.701, attributing a +0.10 gain to a branch that adds just 0.55 M parameters.

The structural and sequence views diverge sharply on individual factors. For the three-zinc-finger protein Egr1 (Zif268; PDB 1a1g), whose 10-bp GC-rich site is read finger-by-finger through recognition-helix residues at positions -1 , $+3$, and $+6$ of each finger [9, 10], TFScope reconstructed all three GCG subsites from sequence alone (oracle $r=0.990$), whereas DeepPBS, operating on the same crystal complex, recovered essentially only the first guanine ($r=0.776$; Fig. 2). That a sequence model recovers the canonical zinc-finger code more faithfully than a structure model reads it from

atoms illustrates how much specificity information the PLM representation already encodes.

These oracle-aligned figures are upper bounds. When the alignment oracle is removed, both methods lose ~ 0.43 in mean r and converge to a statistical tie (TFScope 0.420, DeepPBS 0.419; Table 1). Registration — placing the motif in the correct frame and strand — is thus the dominant residual error for every method evaluated, structure-based included, and is the clearest target for future architectural work rather than a deficiency specific to TFScope.

Retrieval augmentation emulates homology transfer

A central question is whether TFScope’s advantage rests entirely on its retrieval branch. It does not. With retrieval disabled, the sequence encoder alone reached mean $r=0.749$ on the 130-TF split — statistically indistinguishable from DeepPBS (0.750) and already ahead of it on top-1 accuracy (0.797 vs. 0.793) and Matthews correlation (0.699 vs. 0.698; Fig. 3, Table 2). In other words, a model that has never seen a single atomic coordinate matches the structure-based state of the art from sequence alone. Re-enabling retrieval then extends this parity into a clear lead across all base-composition metrics, raising mean r to 0.802 (+0.053).

The retrieval branch behaves as a learned homology-transfer operator: when an experimentally characterized relative of the query exists in the reference set, its PWM supplies a strong prior that the decoder weighs against its own sequence-derived estimate, much as a structural biologist would transfer specificity by homology. This interpretation makes leakage hygiene essential — building the index without leave-gene-out exclusion permits a TF to retrieve a same-gene donor and inflates performance spuriously; all reported figures use the LGO index.

TFScope generalizes to sequence-distant transcription factors

Because near-homologues of test TFs may appear in the DeepPBS training data, strong performance on that split could reflect memorization as much as genuine sequence-to-specificity learning. To test out-of-distribution generalization we constructed *cluster40*, partitioning 1,320 unique proteins into 389 clusters by CD-HIT [19] at 40% identity and splitting them, family-stratified, into 2,983 training, 625 validation, and 639 test PWMs such that no test TF shares $>40\%$ identity with any training TF.

Trained from scratch on this split, TFScope reached mean oracle $r=0.535$ (median 0.505) on

the held-out set (Fig. 4; Table 3), with consistent supporting metrics (IC-weighted r 0.553, top-1 accuracy 0.630, AUC 0.797). Performance varied strongly by family, from bZIP and Homeodomain ($r \approx 0.72$ and 0.68) — compact domains with a single dominant recognition mode — down to long C2H2 zinc-finger arrays ($r \approx 0.43$), which make up 28% of the test set. We emphasize that this is the only leakage-free cluster40 figure: the same checkpoint scores ~ 0.76 and ~ 0.80 on earlier and DeepPBS-style splits, but both are contaminated by training overlap and we therefore report only the clean 0.535.

Leave-family-out analysis exposes a sequence-only transfer floor

To probe the hardest generalization regime, we held out each of the ten structural families in turn, trained on the remaining nine, and evaluated transfer to the unseen family — feasible only because TFScope’s family embeddings are semantic rather than label-derived, so a held-out family still receives a sensible prior. The macro-average oracle r across 4,241 held-out TFs was **0.479 ± 0.057** (median 0.447; Table 4), defining a sequence-only transfer floor to a structurally novel family.

Comparing this floor with in-distribution accuracy is revealing (Fig. 5). The families that performed best in-distribution collapsed the most when held out: bZIP fell by 0.235 (0.72 $\rightarrow$ 0.485) and Homeodomain by 0.202 (0.68 $\rightarrow$ 0.478), indicating that much of their apparent accuracy reflected *family memorization* of a conserved binding grammar rather than transferable sequence-to-specificity mapping. Conversely, families that were weak in-distribution — short C2H2 fingers and ETS — held or even improved out-of-family, consistent with their specificity being driven by local, transferable residue features. The broad in-distribution spread (0.39–0.72) thus compresses to a tight transfer band (~ 0.42 –0.61): once homology is removed, all families converge toward the same ~ 0.48 floor, which is the quantity future architectures must move.

Per-finger recognition codes define the C2H2 prediction frontier

The leave-family-out results single out long C2H2 zinc-finger arrays as the frontier: they showed both the lowest mean transfer r (0.447) and the widest within-family spread of any class (-0.302 to 0.997 across 812 TFs; Fig. 5). The biology explains the spread. A tandem array can read DNA in two qualitatively different ways: through a short recognition motif repeated across fingers, or through a long, finger-specific code in which each finger contacts a distinct triplet [10, 16]. TFScope

predicts the former accurately and the latter poorly. ZNF76, whose recognition is dominated by a short, repetitive finger grammar, was reconstructed almost perfectly even with its entire family held out (oracle $r=0.997$), while ZNF649, a long array with a complex per-finger code and an unusual AT-rich target (ATATAT), collapsed to near-random output ($r=-0.302$; Fig. 6). The discriminating variable is therefore per-finger compositional diversity: where specificity is governed by a single shared grammar, TFScope transfers it across the family boundary; where each finger reads an independent sub-code, the family-level prior offers no traction. This points directly to a finger-resolved model that composes specificity from per-finger contributions as the natural next step for this class.

Family-conditioned experts specialize by structural class

The family-conditioned mixture-of-experts decoder is central to TFScope’s ability to give an unseen family a sensible prior, and we asked whether its routing in fact organizes by structural class or merely distributes load uniformly. Capturing the top-2 routed expert assignments for all 639 held-out cluster40 TFs revealed clearly family-structured routing (Fig. 7): the mean per-family routing entropy was 2.23 bits, well below the 3.58-bit uniform maximum, so each family concentrates its computation on a specialized subset of experts rather than spreading evenly. Several experts emerged as near-pure specialists — one expert handled C2H2_{long} inputs almost exclusively (98.7% of its routed mass), and a cluster of three experts was C2H2_{long}-dominated (59–99%), while a separate expert specialized in Homeodomains (67%). The remaining experts behaved as higher-usage generalists shared across the compact single-consensus families (bZIP, Forkhead, ETS), each of which nonetheless had a clear dominant expert. Notably, the hardest family — long C2H2 zinc-finger arrays — commanded the largest dedicated expert block, indicating that the architecture allocates capacity toward the very class identified above as the prediction frontier. Because this routing structure was measured on out-of-distribution TFs, the specialization generalizes beyond the training set and is not an artifact of memorized identities.

Cross-attention recovers recognition codes from sequence alone

Beyond accuracy, the contact-aware head provides a window onto *which* residues drive each prediction. In its absence, the cross-attention collapses to a degenerate rank-1 pattern: every PWM column

attends to the same two or three residues, with a terminal-residue sink and essentially zero weight on the residues known to read DNA (for KLF4, row-constancy 0.81 and zero mass on the recognition lysine). Activating the contact branch redistributes attention across the domain (Fig. 8). For KLF4 — a Krüppel-like factor whose three C-terminal zinc fingers discriminate GC-box from CACCC elements — attention mass on the recognition-helix lysine K409 rose from 0.000 to 0.094, row-constancy fell from 0.81 to 0.25, and attention entropy rose from 1.47 to 2.36 nats, indicating broad, position-specific attention concentrated near base-contacting residues. The same repair held for the basic-helix-loop-helix factor MyoD, whose basic helix reads the E-box CAGCTG [20]: row-constancy at the recognition leucine L122 fell from 0.59 to 0.27. That an unsupervised attention mechanism, given no structural labels, concentrates on precisely the residues that crystallography identifies as base-contacting demonstrates that TFScope’s predictions are grounded in the correct recognition chemistry. We note one honest limitation: although attention relocates to the causal residues, the present model’s *output* is still largely insensitive to their mutation (wild-type vs. point-mutant PWM $r \approx 0.9997$), because output mutation sensitivity additionally requires a contrastive training signal that we have not yet imposed.

Sequence-only annotation of poorly-characterized transcription factors

The practical promise of a structure-free model is that it can assign binding specificities to factors that have neither a solved complex nor prior experimental characterization. To test this prospectively, we turned to the Codebook/GRECO-BIT consortium, which recently determined binding motifs for hundreds of poorly studied human TFs using GHT- and HT-SELEX and released a set of expert-curated experimental motifs [21]. These motifs constitute a fully independent reference, generated after and apart from our training data. We restricted the comparison to a leakage-controlled subset: the 48 Codebook TFs that fall in our cluster40 held-out partition (held out at 40% identity, never seen by the model), the majority of them zinc-finger proteins from the dark end of the regulatory proteome.

On these 48 TFs, TFScope’s sequence-only predictions matched the independent Codebook motifs at mean $r=0.43$ (median 0.41), with a quarter of the factors predicted at $r > 0.5$ (Fig. 9). The accuracy is bounded by an intrinsic ceiling: our own database targets agree with the Codebook re-measurements at median $r=0.91$, not 1.0, so even a perfect predictor would not reach unity,

and TFScope recovers roughly half of the achievable signal on this hardest, zinc-finger-dominated class — consistent with the C2H2 frontier identified above. Individual predictions are nonetheless strong and useful: the dark zinc-finger factors ZNF516 ($r=0.85$) and ZNF395 ($r=0.75$), the disease-associated SETBP1 ($r=0.74$), the homeobox factor PAX7 ($r=0.64$), and the pluripotency factor POU5F1/OCT4 ($r=0.85$) were all reconstructed faithfully from sequence alone (Fig. 9). TFScope thus provides credible first-pass specificity annotations for factors that are out of reach for structure-based methods, with the zinc-finger frontier setting the honest upper bound on how far sequence-only annotation currently extends.

Discussion

TFScope shows that the binding-specificity information latent in a protein language model, marshaled by architectural components for DBD pooling, family conditioning, contact-aware attention, and homology retrieval, is sufficient to match or exceed the structure-based state of the art for PWM prediction — without any structural input. Five conclusions follow.

First, sequence alone is enough to reach the structural frontier. With retrieval disabled TFScope already ties DeepPBS (mean r 0.749 vs. 0.750) and exceeds it on top-1 accuracy and Matthews correlation; with retrieval it leads on 8 of 11 metrics. Because TFScope requires only the amino-acid sequence, it can be applied to all ~1,600 human TFs and to orthologues, paralogues, and disease variants across the eukaryotic proteome at negligible cost — precisely the regime where docked complexes are unavailable. Validation against the independent Codebook experimental motifs [21] confirms that this reach is real: on held-out, sequence-distant factors the model recovers usable specificities for a meaningful fraction of the dark proteome, with the zinc-finger class marking the current ceiling.

Second, registration is the dominant remaining error. Both TFScope and DeepPBS shed ~0.43 in mean r when motif placement is no longer granted by an oracle. Improving where a motif is placed, rather than which bases it contains, is therefore the highest-value next step, and motivates an explicit placement or gating head.

Third, retrieval is a learned homology-transfer operator, not a crutch. Its margin is real but bounded, and it rests on a sequence encoder that is independently competitive; rigorous leave-gene-

out indexing is essential to measuring it honestly.

Fourth, in-distribution accuracy systematically overstates transfer. Leave-family-out analysis reveals a ~ 0.48 sequence-only transfer floor and shows that the best-performing families in-distribution (bZIP, Homeodomain) owe much of their accuracy to memorized family grammars. Closing the gap between this floor and the in-distribution ceiling is the central open problem, and motivates anti-collapse contrastive objectives [22] and large-scale protein–DNA pretraining.

Fifth, long C2H2 zinc-finger arrays are the frontier. Their per-finger recognition codes violate the one-family-one-consensus assumption behind the family-conditioned prior, producing the widest accuracy spread of any class. Consistent with this, the mixture-of-experts decoder allocates its largest dedicated expert block to this family, yet single shared experts cannot capture per-finger codes that differ from one array to the next; a finger-resolved architecture that composes specificity from individual fingers is the natural remedy.

TFScope has clear limitations. It predicts first-order PWMs and does not model higher-order dependencies, cooperative or heterodimeric binding, or cofactor and chromatin context; its attention localizes causal residues but its output is not yet mutation-sensitive; and very long multi-domain proteins exceed its context window. Each of these defines a concrete direction. Even so, by decoupling specificity prediction from structure determination, TFScope makes proteome-scale, hypothesis- generating annotation of TF binding preferences practical, and provides an interpretable substrate on which these extensions can be built.

Methods

Data and preprocessing

Training data were assembled from the DeepPBS dataset [12] (520 TF–DNA crystal structures with validated PWMs) augmented with CIS-BP [23], RCADE, and MEME-derived entries, yielding 4,247 unique TF records. Each record provides the amino-acid sequence, annotated DBD boundaries, a family label among ten classes (bZIP, Homeodomain, Nuclear_Receptor, Forkhead, bHLH, C2H2_short/medium/long, ETS, Other), and a $4 \times L$ PWM ($L \leq 20$). Each PWM was trimmed to its $IC \geq 0.25$ core; strand was canonicalized by selecting the orientation whose 5' end is more GC-rich, matching the CIS-BP/JASPAR [24] convention.

Model architecture

The encoder is ESM-2 (esm2_t33_650M_UR50D) [14], frozen, with LoRA adapters (rank 16, α 32) on the last six layers and a learned 16-dimensional DBD indicator embedding added within the DBD. Two gated attention-pooling modules (global and DBD streams) yield 1280-dimensional vectors that are concatenated and projected by Linear(2560 $\rightarrow$ 512)+LayerNorm+GELU+Dropout(0.1). The decoder is a DeepSeek-MoE-style block [18] (12 routed + 1 shared experts, top-2 routing, load-balance weight 0.01) conditioned on 2304-dimensional semantic family embeddings. The contact-aware PWM head freezes the legacy cross-attention as a prior and adds a residual branch $z = z_{\text{prior}} + \lambda \Delta z_{\text{contact}}$, where the contact branch uses cosine cross-attention, LayerNorm on keys/values, amino-acid-identity values, and row-diversity and hub penalties to prevent attention collapse; λ is a learned scalar and only the 0.55 M contact-branch parameters are trained. A two-layer MLP position gate emits per-column activation probabilities, and a convolutional regression head emits (4×20) logits passed through a per-column softmax. The retrieval branch injects $K=3$ nearest neighbors (cosine similarity of the ESM-2 DBD representation) through a cross-attention layer before the decoder; the LGO index excludes any neighbor sharing the query’s gene symbol.

Training

Models were optimized with AdamW (learning rate 6×10^{-4} , weight decay 0.01, $\beta_1=0.9$, $\beta_2=0.999$), 500-step warmup and cosine decay, batch size 128, and maximum sequence length 1024. The composite loss combined IC-weighted MSE and per-column cross-entropy for PWM reconstruction, binary cross-entropy on the $\text{IC} \geq 0.25$ indicator for gate supervision, the MoE load-balance term, and an optional in-batch contrastive PWM loss (weight 0.3, temperature 0.1). Training ran on NVIDIA A100 (80 GB) GPUs with early stopping on validation oracle r . The DeepPBS-split model trained only the contact branch on 520 TFs; the cluster40 model trained the full architecture from scratch on 2,983 TFs (best epoch 125); the ten leave-family-out models each trained up to 200 epochs with one family held out, using that family’s correct semantic embedding so the prior carries no label information.

Benchmarks

cluster40 was built with CD-HIT [19] at 40% identity over 1,320 proteins (389 clusters), family-stratified 2,983/625/639. *Leave-family-out* splits held out each of the ten families in turn, drawing validation from the remaining families by 40%-identity clustering. DeepPBS predictions were read from the published blind-benchmark output ($n=130$) and passed through the identical trimmed-core oracle aligner; for the deployable protocol, both methods' outputs were canonicalized identically.

Evaluation

The primary metric is per-TF oracle Pearson r : gate-active predicted columns are aligned to the $IC \geq 0.25$ target core (offset ± 10 , reverse-complement) and r is averaged column-wise. The supporting panel adds IC-weighted r , MAE, RMSE, cross-entropy, KL divergence, top-1 (argmax) accuracy, macro AUC, macro F1, and MCC. For the deployable protocol, prediction and target are canonicalized independently and scored without alignment.

Attention analysis

KLF4 and MyoD DBD sequences were run with retrieval disabled through the model with and without the contact branch. Row-constancy is the mean off-diagonal correlation among attention rows; entropy is the Shannon entropy of the column-averaged attention distribution; residue mass is the summed attention at the annotated recognition residue. Output mutation sensitivity was assessed by comparing wild-type and single-point-mutant output PWMs.

Data availability

DeepPBS benchmark data are available from the original publication [12]. CIS-BP [23] and JASPAR [24] data are available at <http://cisbp.ccbbr.utoronto.ca/> and <http://jaspar.genereg.net/>. The augmented training dataset and all splits will be deposited to Zenodo upon acceptance.

Code availability

TFScope source code and trained checkpoints will be released publicly upon acceptance.

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Author contributions

L.H. conceived the study, designed and implemented TFScope, performed all experiments, and wrote the manuscript. [Co-author contributions TBD.]

Competing interests

The authors declare no competing interests.

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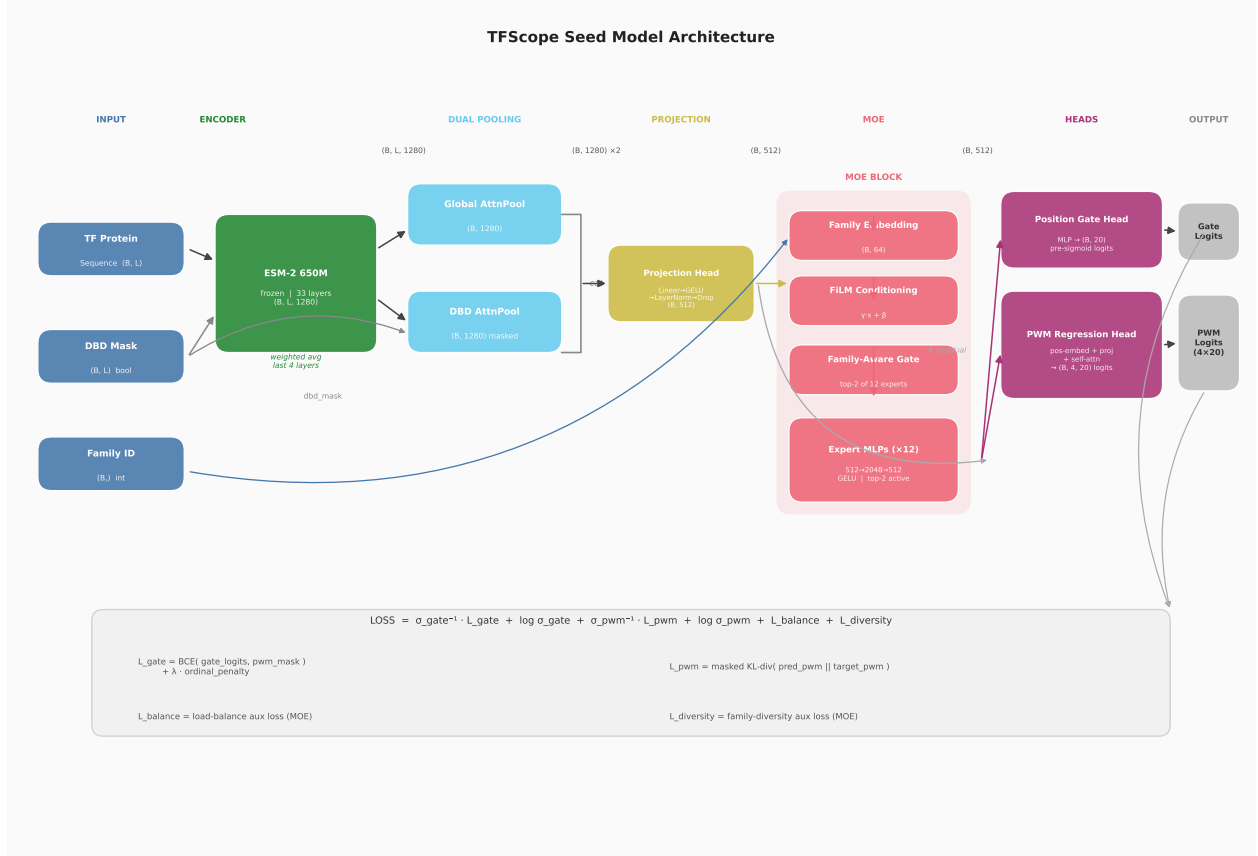


Fig. 1: The TFScope framework. A TF amino-acid sequence passes through (i) a frozen ESM-2 encoder with LoRA adapters and a DBD indicator embedding; (ii) dual-stream gated pooling over the full sequence and the DBD; (iii) a family-conditioned mixture-of-experts decoder (12 routed + 1 shared expert, top-2 routing) driven by semantic family embeddings; (iv) a contact-aware cross-attention head that augments a frozen prior with a λ -weighted residual branch; and (v) a position gate and PWM regression head. An optional leave-gene-out retrieval branch injects $K=3$ nearest-neighbor PWMs before the decoder. Only ~ 0.55 M parameters (contact branch) are trained on the DeepPBS split; the encoder and prior remain frozen.

TFScope (sequence only) predicts the Egr1 binding motif better than structure-based DeepPBS on the same crystal complex

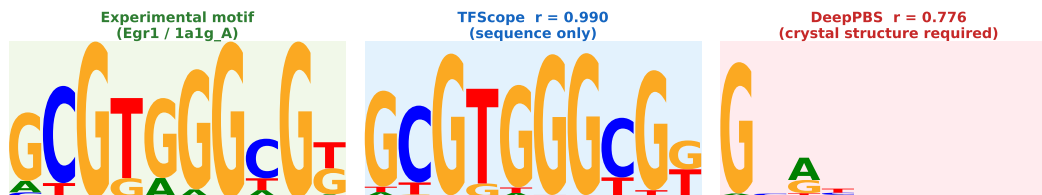


Fig. 2: Sequence beats structure on the zinc-finger code (Egr1). Sequence logos for Egr1/Zif268 (PDB 1a1g, chain A): the experimental motif (JASPAR MA0162.1; top), the TFScope prediction from sequence alone (oracle $r=0.990$; middle), and the DeepPBS prediction from the crystal complex (oracle $r=0.776$; bottom). Egr1 reads its 10-bp GC-rich site finger-by-finger via recognition-helix residues [9]; TFScope reconstructs all three GCG subsites, whereas the structure-based prediction is nearly flat.

Retrieval augmentation (RAG) drives TFScope above the structure-based baseline
Without RAG, TFScope ties DeepPBS; with RAG it exceeds on 4/6 metrics shown

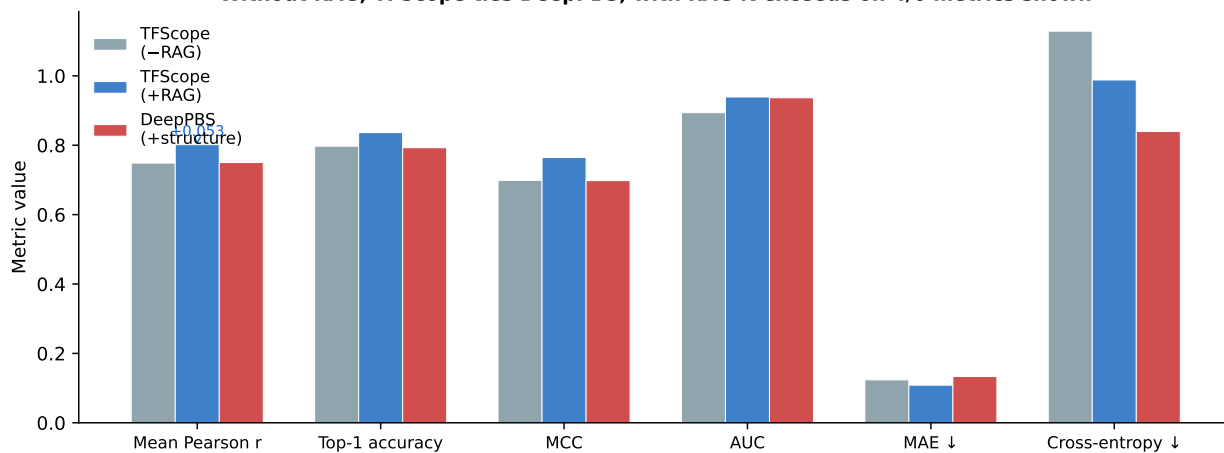
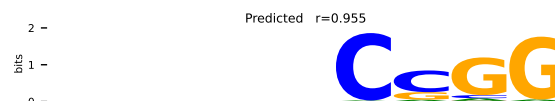
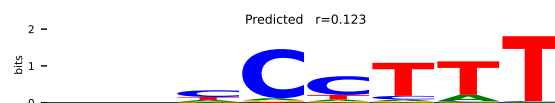
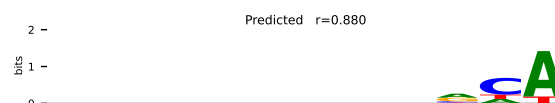
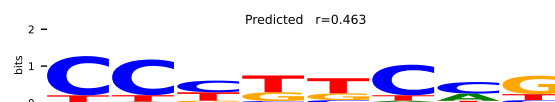
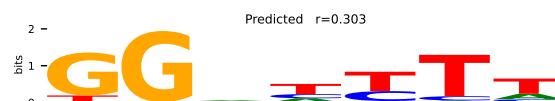
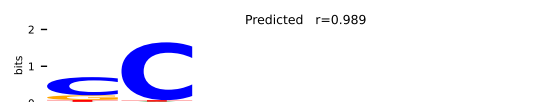
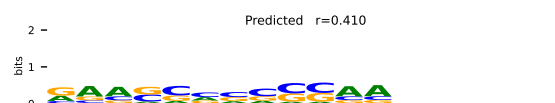
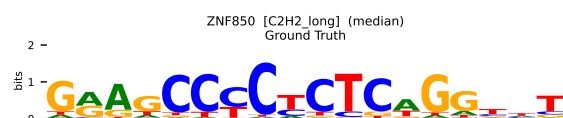
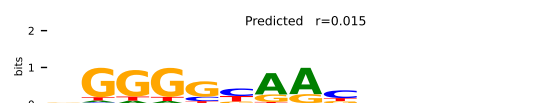


Fig. 3: The sequence encoder alone matches DeepPBS; retrieval extends the lead. Six motif-level metrics on the 130-TF DeepPBS blind split for TFScope without retrieval (−RAG), TFScope with retrieval, and DeepPBS. Without retrieval, TFScope ties DeepPBS on mean r (0.749 vs. 0.750) and already exceeds it on top-1 accuracy and MCC; retrieval raises mean r to 0.802. DeepPBS retains an advantage only on calibration (cross-entropy, KL).

Ground Truth

Prediction

TFScope (c40_best) — Predicted vs Ground Truth PWMs
(cluster40 test set, oracle-aligned, IC $\geq$ 0.25 bits)



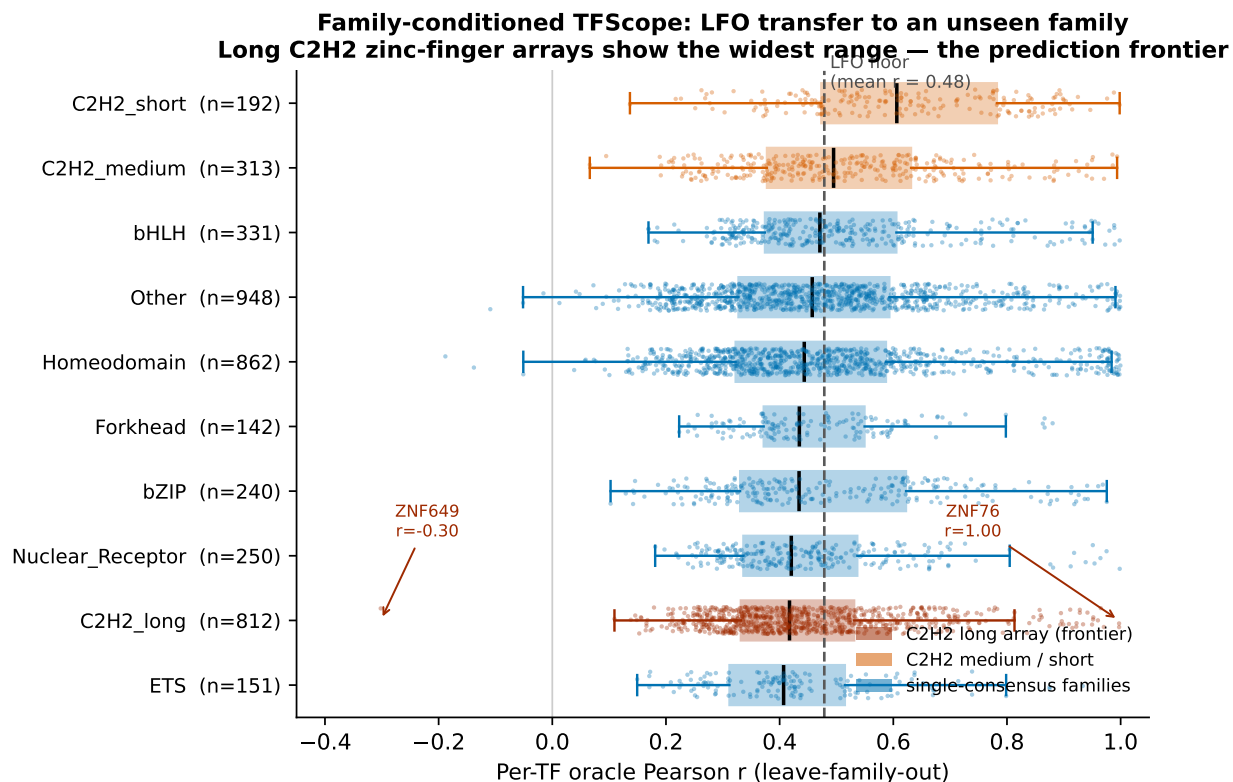


Fig. 5: A sequence-only transfer floor and the C2H2 frontier (leave-family-out). Per-TF oracle r under leave-family-out for all ten families (4,241 TFs), ordered by median. The dashed line marks the macro-average transfer floor ($r=0.479$). The broad in-distribution spread compresses to a tight transfer band; long C2H2 arrays (darker) show the lowest mean and the widest spread, marking them as the prediction frontier.

Same C2H2_long family — opposite LFO outcomes
Simple 2-finger recognition (ZNF76) transfers; complex long-array grammar (ZNF649) does not

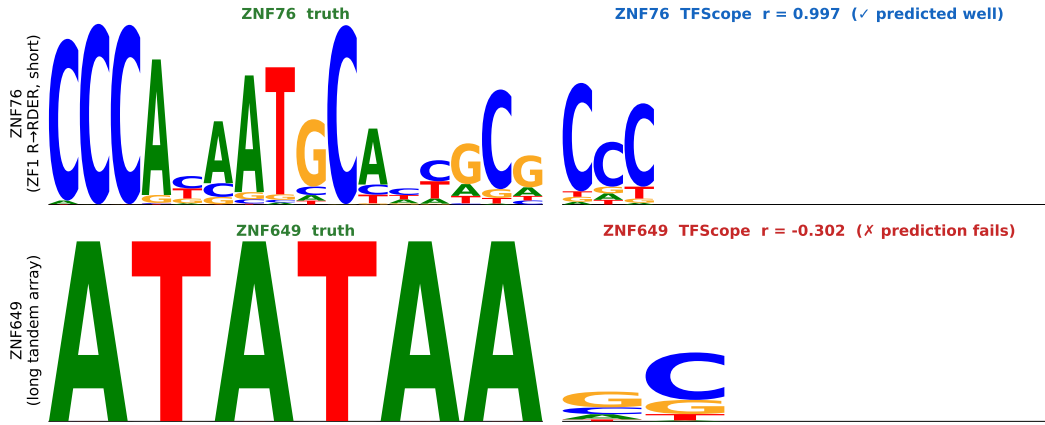


Fig. 6: Per-finger code complexity decides outcomes in long C2H2 arrays. Experimental (top) and TFSScope-predicted (bottom) logos for the best- and worst-transferring C2H2_long TFs under leave-family-out. ZNF76, dominated by a short repetitive finger grammar, is reconstructed almost perfectly (oracle $r=0.997$); ZNF649, a long array with a complex per-finger code and an AT-rich target, collapses to near-random ($r=-0.302$). Both had their entire family held out.

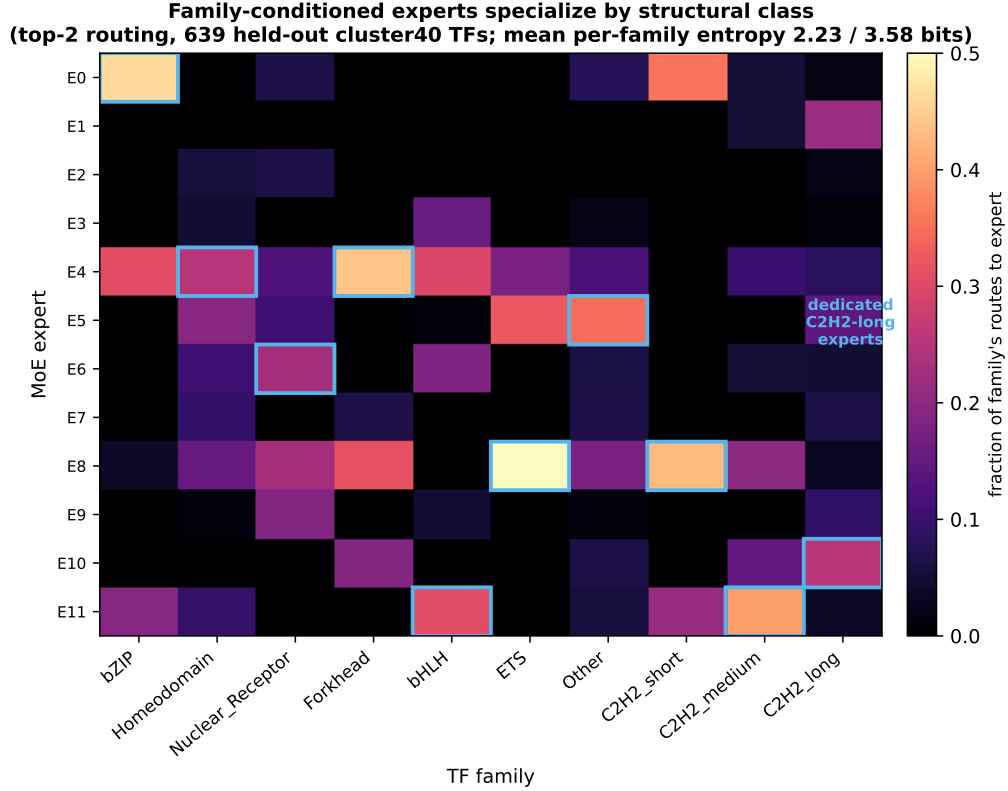


Fig. 7: Family-conditioned experts specialize by structural class. Column-normalized top-2 routing frequency of each TF family (columns) over the twelve routed experts (rows), measured on the 639 held-out cluster40 TFs. Cyan boxes mark each family's dominant expert. Routing is strongly family-structured (mean per-family entropy 2.23 of a possible 3.58 bits): one expert is a near-pure C2H2_long specialist (98.7%) and a cluster of three experts is C2H2_long-dominated, giving the prediction-frontier family the largest dedicated capacity, while a separate expert specializes in Homeodomains. Compact single-consensus families share higher-usage generalist experts. Specialization measured on out-of-distribution TFs indicates it generalizes beyond training identities.

TFScope cross-attention: degenerate collapse repaired by the contact branch
Cyan dashed line = known specificity-determining residue

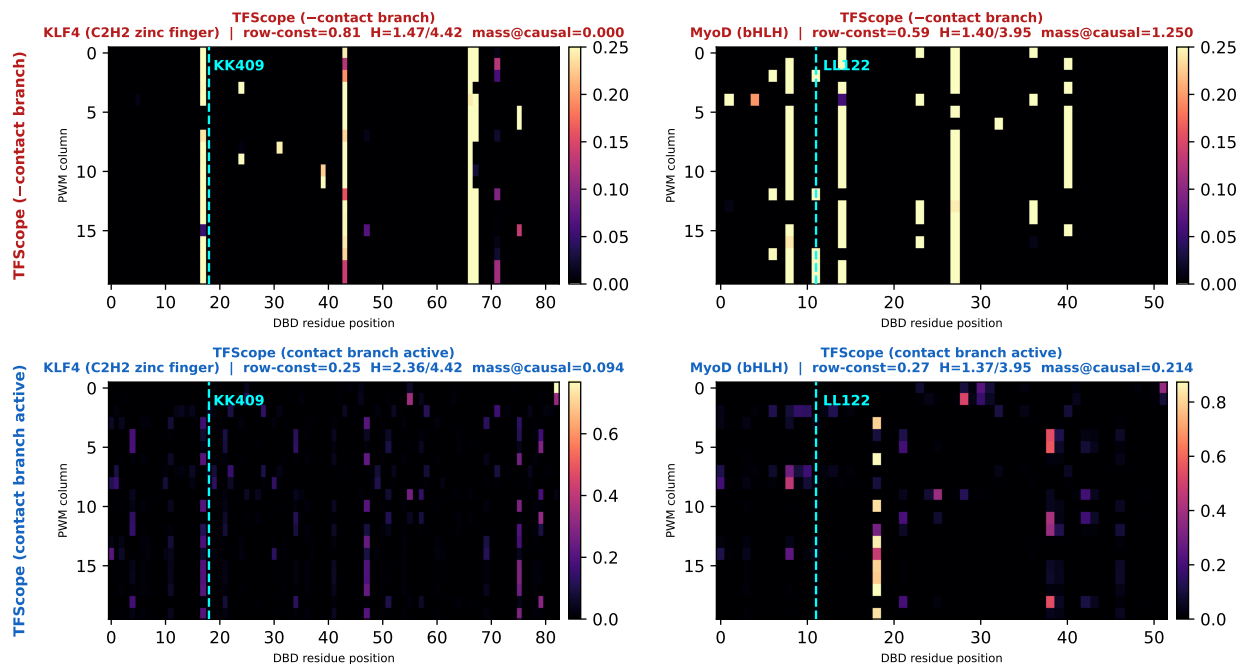
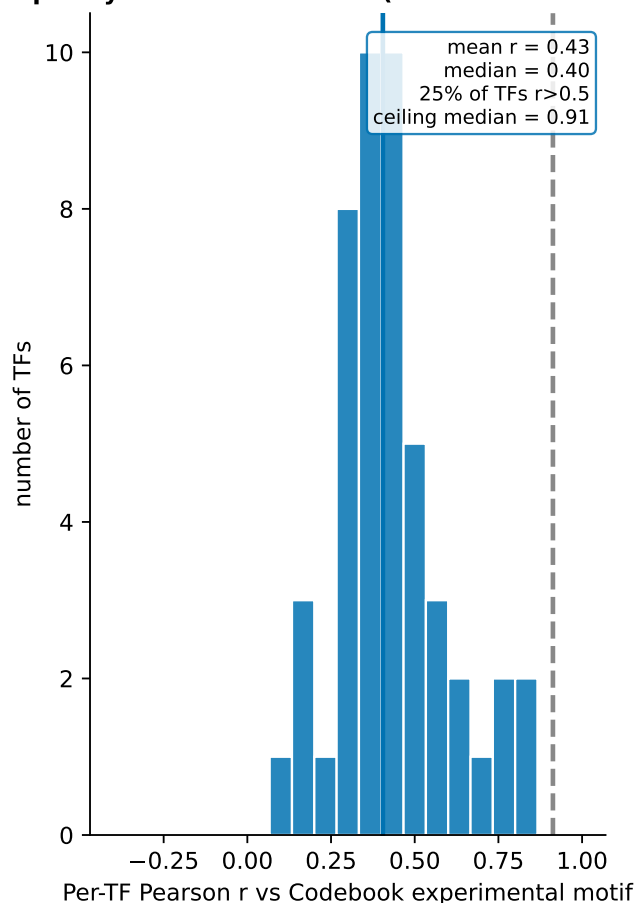


Fig. 8: Cross-attention concentrates on base-contacting residues. Attention from each PWM column (y) to each DBD residue (x) for KLF4 (C2H2) and MyoD (bHLH), without (top) and with (bottom) the contact-aware branch. The cyan line marks the experimentally identified recognition residue (KLF4 K409; MyoD L122). The contact branch breaks the degenerate rank-1 pattern (KLF4 row-constancy 0.81 $\rightarrow$ 0.25) and directs attention onto the recognition residue, recovering the correct recognition chemistry without structural supervision.

TFScope provides sequence-only specificity for poorly-characterized TFs

Sequence-only annotation of 48 poorly-characterized TFs (external validation)



held out at 40% identity;
blue line = TFScope, grey = achievable ceiling

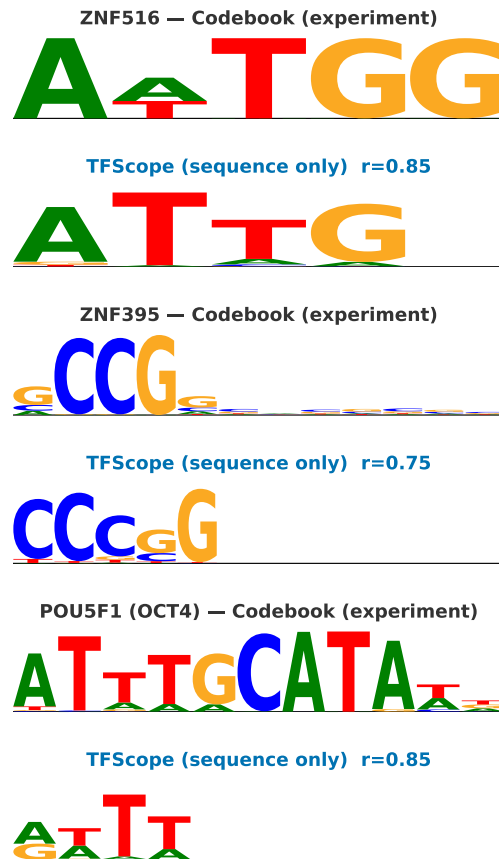


Fig. 9: Sequence-only annotation of poorly-characterized TFs, validated against external Codebook motifs. **Left**, distribution of per-TF Pearson r between TFScope’s sequence-only prediction and the independent Codebook expert-curated experimental motif [21], for the 48 Codebook TFs in our cluster40 held-out partition (held out at 40% identity; never seen by the model). Blue line, TFScope median; grey dashed line, the database-concordance ceiling (our own database targets vs. Codebook), which bounds achievable accuracy. **Right**, example motifs (Codebook experimental, top; TFScope prediction, bottom) for two dark zinc-finger factors (ZNF516, ZNF395) and the pluripotency factor POU5F1/OCT4, all reconstructed from protein sequence alone. The set is zinc-finger-dominated, placing it at the model’s honest frontier; nonetheless a substantial fraction of these otherwise-uncharacterized factors receive usable specificity annotations.

Table 1: Motif-level accuracy on the DeepPBS 130-TF blind split. Oracle-aligned protocol ($IC \geq 0.25$ core, offset ± 10 and reverse-complement alignment), $n=116$ TFs covered by all methods. Bold marks the better of TFScope vs. DeepPBS. For CE and KL, lower is better. The final two rows give the deployable canonical-fixed mean r (no alignment oracle), where the two methods tie.

Metric	TFScope (–contact)	TFScope	DeepPBS
Mean Pearson r	0.701	0.802	0.750
Median Pearson r	0.723	0.831	0.759
Top-1 accuracy	0.753	0.836	0.793
MCC	0.615	0.765	0.698
F1 score	0.646	0.765	0.724
AUC	0.879	0.939	0.937
IC-weighted r	0.954	0.972	0.968
MAE	0.141	0.108	0.133
RMSE	0.242	0.184	0.204
Cross-entropy (CE)	1.140	0.988	0.840
KL divergence	0.666	0.517	0.358
<i>Deployable (canonical-fixed, no alignment oracle)</i>			
Mean Pearson r	—	0.420	0.419

Table 2: Retrieval ablation on the DeepPBS blind split. With retrieval disabled, the TFScope sequence encoder alone already matches DeepPBS on mean r and exceeds it on top-1 accuracy and MCC; retrieval extends the margin across all base-composition metrics.

Metric	TFScope –RAG	TFScope +RAG	DeepPBS
Mean Pearson r	0.749	0.802	0.750
Top-1 accuracy	0.797	0.836	0.793
MCC	0.699	0.765	0.698
F1 score	0.696	0.765	0.724
MAE	0.124	0.108	0.133
Cross-entropy	1.129	0.988	0.840

Table 3: Out-of-distribution accuracy on cluster40. 639-TF held-out set; no test TF within 40% identity of any training TF. Model trained from scratch on 2,983 cluster40 training TFs.

Metric	TFScope (cluster40)
Mean oracle r	0.535
Median oracle r	0.505
IC-weighted r	0.553
Top-1 accuracy	0.630
AUC	0.797
F1 score	0.603
MCC	0.472
MAE	0.198

Table 4: Leave-family-out transfer by family. Each model was trained with the named family entirely held out and evaluated on that family; values are reproduced from per-TF scores. Families are ordered by mean transfer r . The Δ column gives the change from the same family’s cluster40 in-distribution score where available, capturing family memorization; large negative values indicate accuracy that does not transfer.

Family	n	Mean r	Median r	Δ in-dist
ETS	151	0.424	0.407	—
C2H2_long	812	0.447	0.417	+0.017
Nuclear_Receptor	250	0.454	0.421	—
Forkhead	142	0.464	0.435	—
Other	948	0.473	0.457	—
Homeodomain	862	0.478	0.443	−0.202
bZIP	240	0.485	0.434	−0.235
bHLH	331	0.508	0.471	+0.008
C2H2_medium	313	0.514	0.495	+0.014
C2H2_short	192	0.613	0.606	+0.223
Macro avg	4241	0.479	0.447	—